HiViG: A History-Aware Visually Grounded Critic for Computer Use Agents

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Paper version: arXiv 2606.11078v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *relying on post-execution correction is often unsafe and impractical, motivating the use of test-time intervention*

Critic models judge the policy's next action, but forget history and never check the pixels.

Motivation: why critics fail

1 **Short-sighted.** No compressed record of past steps; policies re-explore.

2 **Ungrounded.** Critics read the intent, not the coordinates.

3 **Too late.** GUI actions are irreversible; verdicts must precede execution.

Evaluation setup

WALv2 · 154 web · AndroidLab · 138 · WAA · 145 desktop · 32B + G-3-Flash policies · Critic: Qwen3-VL-8B · 15 / 30-step horizons

- **All online**, execution-based success; no offline step metric.
- **Single run** per cell; no seeds, no variance.

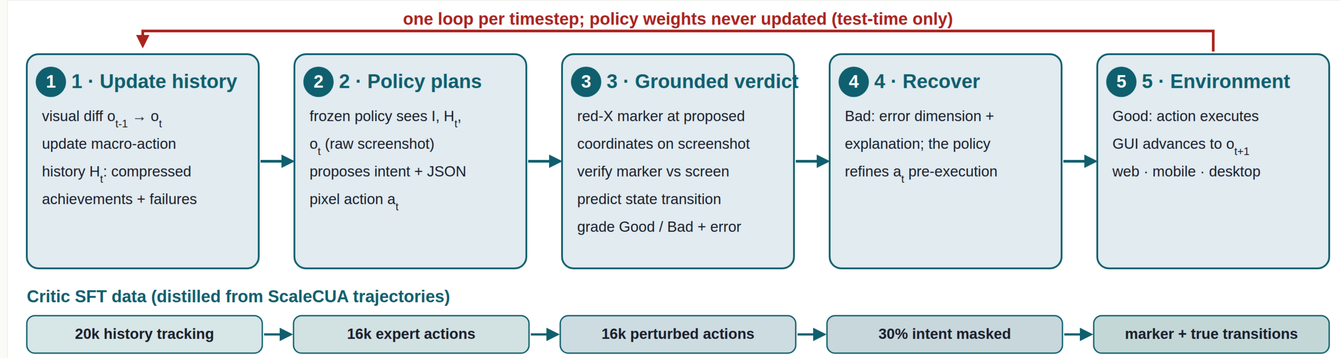
Our read (CUA Radar)

63	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; matched-cost baselines; ablations; cost.
- **Flags:** single run, no variance; critic unvalidated; held-out unclear.
- Pattern C18. Radar's judgment; numbers from the paper.

Method: an 8B critic wrapped around a frozen policy

- **Sees / acts:** screenshots only (no DOM, no a11y tree); JSON pixel actions on a 1000×1000 grid; web + mobile + desktop.
- **Critic:** 8B SFT (52k samples) outputs a macro-action history + a marker-grounded Good/Bad verdict over a 12-error taxonomy.
- **Signal:** pre-execution verdict + explanation; policy refines; weights frozen. Trained on verified state-transitions.



Our redraw (not the paper's figure): one timestep of the HiViG loop — history update, planning, grounded verdict, recovery.

$$a_t \sim \pi(\cdot \mid I, H_{t-1}, o_t)$$

the frozen policy acts on instruction, history, current screenshot

Results: online success rate (%)

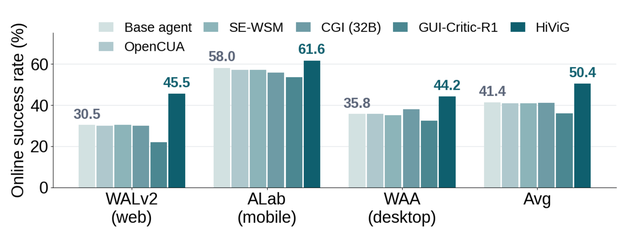


Table 1, Gemini-3-Flash policy, redrawn; execution-based, single run.

WALv2: 30.5 → 45.5 (Gemini-3-Flash); 13.0 → 25.3 (Qwen3-VL-32B)

SO WHAT → *Qwen3-VL-32B-Thinking panel in the controls table.*

Ablation: components on WALv2 (Gemini-3-Flash)

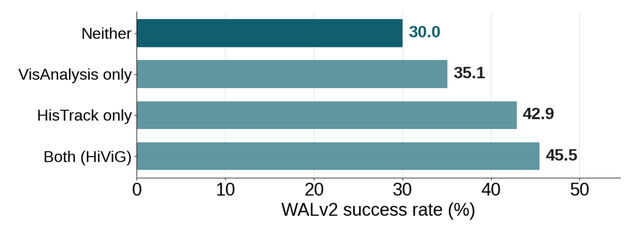


Table 2, redrawn; single run, Gemini-3-Flash policy on WALv2.

SO WHAT → *Either capability alone beats every baseline; both together reach 45.5; a single critic inference per step already suffices. HisTrack alone is the stronger single driver (42.9 vs 35.1). Data ablations (Qwen, WALv2): mixed training 25.3 vs separate critics 23.4; without ground-truth state-transitions 20.8.*

Headline Numbers

45.5%
WALv2 success
Gemini-3-Flash; base 30.5

+9.0 avg vs best basel.

38.3% Qwen-32B avg

52k critic SFT samples

The paper's own controls

CONTROL	RESULT	READING
Qwen-32B policy	avg 31.0 → 38.3	+5.8 vs best
No marker	WALv2 25.3 → 20.8	misreads coords
No intent mask	ALab 51.5 → 47.1	text shortcut
Critic calls/step	2; 1 is enough	one suffices

Takeaway

An 8B critic's marker-grounded pre-execution verdicts + compressed history lift frozen policies by avg +5.8 (Qwen-32B) / +9.0 (Gemini-3-Flash) vs the strongest baseline; single-run, critic unvalidated.

52k SFT samples · 12 error dimensions · code on GitHub

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Paper · Code